

Differences in codon usage between host-species-specific rabies virus clades are driven by UpA and purine content

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Abstract

Viral genes sometimes use certain codons more than others due to their nucleotide content, translational efficiency, and selection pressure from the host immune system. The rabies virus (RABV) is a negative strand RNA virus which can infect a broad range of mammalian hosts, with many of its clades circulating predominantly in specific host species. Previous work on codon usage in RABV has focused only on broader viral clades. We use publicly available RABV nucleoprotein gene sequences to investigate how dinucleotide content and codon usage

biases differ between host-associated clades, and what drives these differences. We found that codon usage varies most between bat- and carnivore-associated RABV clades, and more subtly between host-species-specific minor clades within these groups. Pyrimidine and UpA content were both found to have a strong influence over codon usage patterns, and CpG content was considerably higher in carnivore-associated RABV clades than in bat-associated clades. This, along with a reduced number of zinc-finger antiviral protein binding motifs in bat-associated RABV sequences, suggests that bat-associated RABV clades may be under higher selection pressure from the host's zinc-finger antiviral protein than carnivore-associated clades are, warranting further investigation of the mechanism underpinning this change.

Keywords: lyssavirus, CpG content, zinc-finger antiviral protein, host adaptation, virus-host interactions

Introduction

Despite encoding the same amino acid, synonymous codons are often not used equally. While a wide range of factors likely influence these imbalances (Parvathy *et al.* 2022), codon usage biases in both RNA and DNA viruses generally appear to be shaped by nucleotide-level effects such as guanine and cytosine (GC) content rather than by codon-level translational selection (Jenkins & Holmes 2003; Shackelton *et al.* 2006). Unlike positive-strand RNA viruses (Tian *et al.* 2018), negative-strand RNA viruses do not appear to mimic their host species' codon usage patterns (Rima 2015), despite also being reliant on the host's translation machinery. This may be due to transcription of viral RNA being dependent on RNA stability, which is influenced by codon usage (Gumpper *et al.* 2019). Nonetheless, using machine learning methods, codon usage together with nucleotide and dinucleotide content can accurately predict the hosts of coronaviruses (Brierley & Fowler 2021) and other RNA viruses (both positive- and negative-strand; Babayan *et al.* 2018), suggesting a connection between viral codon usage and host species.

The rabies virus (RABV) is a negative-strand RNA virus that can infect a broad range of mammal species. The RABV genome consists of five genes, which encode the nucleoprotein (N), phosphoprotein (P), matrix protein (M), glycoprotein (G) and RNA polymerase (L; "large" protein). RABV can be broadly split into bat- and carnivore-associated clades, with the main bat-to-carnivore host shift event being estimated to have occurred approximately 600 years ago (Troupin *et al.* 2016). Another shift, from bats to skunks and raccoons, is estimated to have occurred 250 years ago (Ding *et al.* 2017), and bat-to-carnivore spillover events continue to occur in the present day (Kuzmin *et al.* 2012). Some RABV clades appear to be highly host-specific, circulating within a single species; for example, sustained transmission cycles have been observed in Chinese ferret badger (CFB) populations (Zhang *et al.* 2013) associated with

the minor clade Asian SEA2b, as opposed to the clade circulating sympatrically in dogs, Asian SEA2a.

There has been debate as to whether these host-species-specific transmission cycles are due to adaptation or ecological circumstances (Marston *et al.* 2018; Mollentze *et al.* 2014), with evidence of positive selection over host shifts being mixed: some studies show no evidence of adaptation to new host species (Kuzmin *et al.* 2012; Troupin *et al.* 2016), while others observed some positively selected sites but with low repetition across different host shift events (Streicker *et al.* 2012). Troupin *et al.* (2016) also found, however, that an amino acid change from leucine to serine at nucleoprotein position 374 appears ubiquitously, and almost exclusively, in two separately emerging ferret badger-associated clades, which suggests that this amino acid change is required for sustained transmission in ferret badgers. These studies focused on non-synonymous substitutions (changes to the amino acid sequence and therefore the structure of the encoded protein), leaving the possible influence of synonymous substitutions (i.e., changes in codon usage bias or dinucleotide bias) on RABV-host adaptation yet to be investigated.

On average, across all available whole genome sequences, RABV has the most balanced codon usage among lyssaviruses (Zhang *et al.* 2018), perhaps due to its broad host range compared to other members of the family (Marston *et al.* 2018). Despite this, codon selection is important to RABV fitness; viral replication is inhibited when the RABV matrix protein sequence is reconstructed with codons that are rarely used in the cells that the virus is cultured in (Luo *et al.* 2020a). This may be due to a collateral increase in CpG and UpA dinucleotides, which are selected against by host immune mechanisms such as the zinc-finger antiviral protein (ZAP; Tulloch *et al.* 2014). Previous codon usage studies focusing on the rabies virus either did not

investigate differences in codon usage on a clade level (Morla *et al.* 2016), or subdivided the sequences into the major RABV clades (Bat, Africa-2, Arctic, Cosmopolitan, Asian and Indian subcontinent major clades in He *et al.* (2017); plus Africa-3 and South-East Asian clades in Li *et al.* (2023)), potentially losing signal from host-species-specific minor clades. Investigating differences in codon usage between host-species-specific minor clades has the potential to shed light on the effect of sustained transmission within a single host species on codon usage.

In this study we aimed to quantify differences in codon usage biases between host-species-specific RABV minor clades, and investigate what factors underpin these differences. We hypothesised that clades that diverged more deeply in the virus' evolutionary history (i.e., from bats to carnivores) would show clear codon usage biases due to longer periods of sustained transmission within specific host groups, while more recent host shifts (i.e., to Chinese ferret badgers) would show less obvious differences to their most closely related clades.

Methods

Data acquisition

We downloaded complete RABV N gene sequences from the RABV-GLUE sequence database (Campbell *et al.* 2022; acquired 15 December 2023). The N gene was chosen as it was the most commonly sequenced gene across a wide range of clades. We searched for sequences with 100% N gene coverage from a selection of host-species-specific minor clades spanning a wide range of host species, including those mentioned by Troupin *et al.* (2016) and four bat-associated clades. The full list of clades and the terms used to filter sequences on RABV-GLUE are shown in table 1.

Table 1: RABV-GLUE clades and filtering terms used when downloading RABV N gene sequences. Number of sequences and date range include only sequences used in the final dataset.

RABV-GLUE Clade	Host	RABV-GLUE host filter term	No. sequences	Sampling date range
Asian SEA2a	Domestic dog	“Canis familiaris”, “Canis lupus familiaris”	76	1994-2015
Asian SEA2b	Chinese ferret badger	“Melogale moschata”, “Chinese ferret badger”	35	2008-2013
Arctic A	Arctic fox	“Vulpes lagopus”	17	1977-2017
Cosmopolitan AM2a	Mongoose	“Mongoose”	9	2000-2002
Cosmopolitan AF1b	Domestic dog	“Canis familiaris”, “Canis lupus familiaris”	161	1986-2017
RAC-SK	Raccoon	“Procyon lotor”	N/A	N/A
RAC-SK	Skunk	“Mephitis”, “skunk”	9	2009-2017
Bat DR	Vampire bat	“Desmodus rotundus”	39	1988-2018
Bat TB1	Mexican free-tailed bat	“Tadarida brasiliensis”	32	2002-2005
Bat EF-E2	Big brown bat	“Eptesicus fuscus”	36	2003-2005
Bat LC	Hoary bat	“Lasiurus cinereus”	16	1998-2005

Raccoon sequences and raccoon-related skunk sequences were removed from the study after initial phylogenetic analysis suggested substantial transmission between raccoons and skunks. The southern central skunk variant (SCSK) within the RAC-SK major clade did not contain any raccoon-derived sequences, suggesting little sampled transmission between the species, and was therefore retained. Five additional sequences from the Arctic A clade were added to

the dataset which had been collected from *Vulpes lagopus*, but mislabelled as originating from *Vulpes vulpes* on RABV-GLUE. Sequences containing ambiguous bases (i.e., N, R or Y) were removed from the dataset, leaving 430 complete N gene sequences in total. The GenBank accession numbers for these sequences are available in Supplementary Table S1.

Phylogenetic trees

Phylogenetic trees were constructed using IQtree (Nguyen *et al.* 2015) with a Gannoruwa bat lyssavirus (GBLV) sequence (accession number NC_031988) used as an outgroup, ultrafast bootstrapping with 1000 replicates, and the best-fitting model determined by IQTree's ModelFinder (GTR+F+I+G4). Visualisation was carried out in R version 4.5.1 (R Core Team 2020) using the *ggtree* R package (Yu *et al.* 2017).

Codon usage

We used the effective number of codons (ENC; a measure of overall codon usage bias accounting for amino acid composition, with values ranging from 20 to 61, where lower values correspond to stricter biases) to compare the strength of codon usage biases between the RABV clades, calculated using the equations shown in Wright (1990):

$$\hat{F} = \frac{n \sum_{i=1}^k p_i^2 - 1}{n - 1}$$

where $\hat{F}$ is the homozygosity of each amino acid, p_i is the frequency of the synonymous codon i , k is the number of codons that encode the amino acid, and n is the total frequency of the amino acid; and:

$$ENC = 2 + \frac{9}{\overline{F_2}} + \frac{1}{\overline{F_3}} + \frac{5}{\overline{F_4}} + \frac{3}{\overline{F_6}}$$

where $\overline{F_x}$ is the mean homozygosity of the group of amino acids encoded by x synonymous codons.

Raw codon usage and relative synonymous codon usage (RSCU) values were calculated for each minor clade using CAIcal (Puigbò *et al.* 2008a), which uses the following formula (Sharp & Li 1986):

$$RSCU_{i,j} = \frac{n_i X_{i,j}}{\sum_{j=1}^{n_i} X_{i,j}}$$

where n_i is the number of synonymous codons for amino acid j , and $X_{i,j}$ is the number of occurrences of codon i for amino acid j . The raw codon usage counts (excluding amino acids with only one synonymous codon) were analysed using principal component analysis (PCA) using the *stats* R package (R Core Team 2020).

Codon adaptation index (CAI) values were calculated for each RABV clade against four host species using CAIcal to measure the similarity in codon usage between each RABV clade and the corresponding host species. CAI is calculated from a viral sequence and a host reference codon usage table with the following equation (Sharp & Li 1987):

$$CAI = \frac{\prod_{k=1}^L RSCU_k^{\frac{1}{L}}}{\prod_{k=1}^L RSCU_{kmax}^{\frac{1}{L}}}$$

where for each codon k in the viral sequence, $RSCU_k$ is the RSCU of codon k in the host reference table, $RSCU_{kmax}$ is the maximum RSCU in the host reference table for the amino acid encoded by codon k , and L is the total number of codons in the viral sequence (excluding

the start codon). The reference host codon usage datasets were obtained for domestic dogs (*Canis familiaris*), Arctic foxes (*Vulpes lagopus*), big brown bats (*Eptesicus fuscus*) and vampire bats (*Desmodus rotundus*) from CoCoPUTs (Alexaki *et al.* 2019). Other host species had no available codon usage data, so all 10 RABV clades in our dataset were compared to the above four host species. The CAI values were divided by the expected CAI (eCAI; a 95% upper confidence limit for CAI of a random sequence, controlling for amino acid usage and GC content (Puigbò *et al.* 2008b)) values to give a normalised CAI (nCAI).

Dinucleotide content

The ratio of observed-to-expected CpG and UpA content for each sequence was calculated as in Gardiner-Garden & Frommer (1987):

$$Obs/Exp \text{ dinucleotide } n_1n_2 = \frac{No. \text{ dinucleotide } n_1n_2}{No. \text{ nucleotide } n_1 \times No. \text{ nucleotide } n_2} \times N$$

where N is the nucleotide length of the sequence, and n_1 and n_2 are the first and second nucleotides in the dinucleotide respectively.

Ancestral sequence reconstruction was carried out using IQtree and the ratios of observed-to-expected CpG and UpA content of the resulting ancestral sequences were calculated to examine when in the rabies virus' evolutionary history changes in CpG and UpA content may have occurred.

We also compared the frequency of ZAP's CpG-based optimal binding motif, C(n_m)G(n)CG (where n can be any nucleotide, and the subscript m denotes the length of this subsequence, being either 4, 5, 6, 7 or 8 nucleotides long, determined in mice by Luo *et al.* (2020b)), to a similar but suboptimal negative control motif, C(n_m)C(n)CG, in the modern N gene sequences

of each clade to test whether the difference in CpG content between carnivore- and bat-associated clades was likely caused by ZAP-based selection. The CpG-based binding motif has been previously analyzed in SARS-CoV-2 (Afrasiabi *et al.* 2022) and hepatitis C virus (Mukherjee *et al.* 2023) sequences, with results suggesting that the optimal motif found in mice is shared by humans; little is known about ZAP binding motifs in other mammalian species. The UpA-based ZAP optimal binding motif is currently unknown, so we could not perform a corresponding UpA-based investigation.

Results

The RABV phylogeny can be broadly split into carnivore-associated (Cosmopolitan AF1b, Cosmopolitan AM2a, Arctic A, Asian SEA2a and Asian SEA2b) and bat-associated (Bat DR, Bat TB1, Bat LC and Bat EF-E2) clades, with the exception of the skunk-associated RAC-SK SCSK clade, although this has lower bootstrap support than the other clades (78%; Figure 1). This clade is thought to have descended from bat-associated clades separately to the main bat-to-carnivore host shift (Ding *et al.* 2017; Troupin *et al.* 2016). The only paraphyletic RABV-GLUE-designated clade was the Asian SEA2a domestic dog-associated clade, which also contains the more recently diverged Asian SEA2b clade, associated with Chinese ferret badgers.

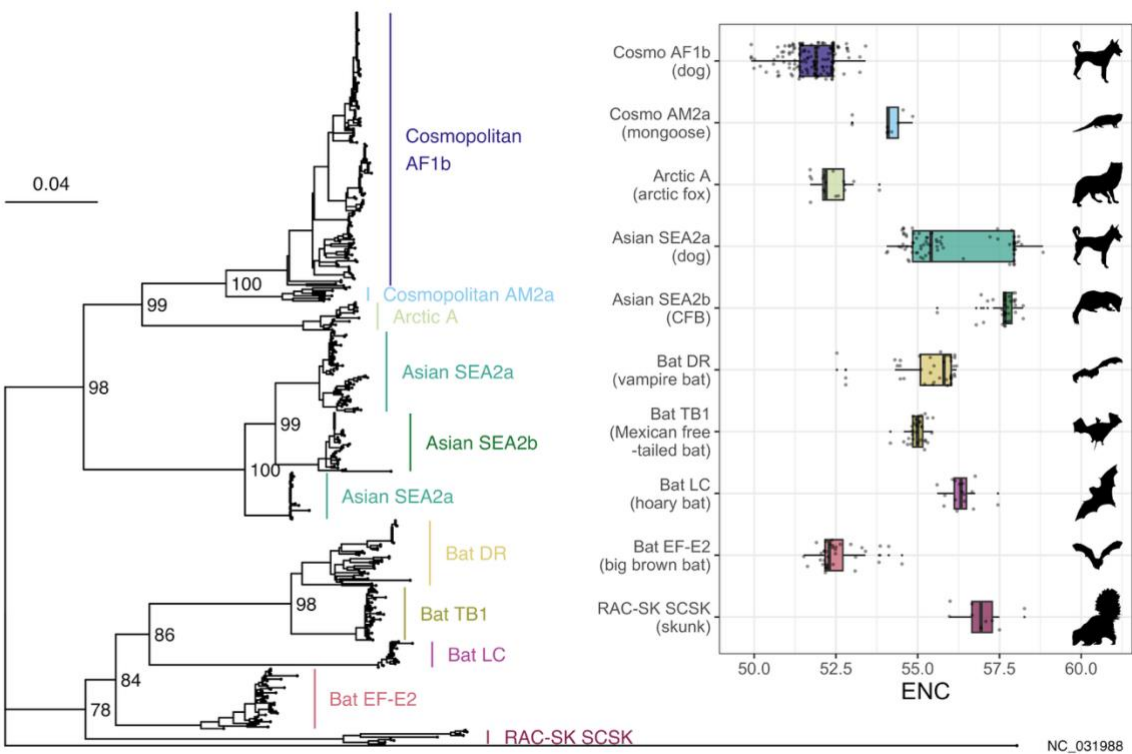


Figure 1: Maximum-likelihood phylogenetic tree of all the sequences used in analyses. A GBLV sequence (NC_031988) was used as an outgroup. Branch lengths are scaled by substitutions per site. Internal node labels are bootstrap support values. **Inset:** Effective number of codons (ENC) across different host-species-specific rabies clades. Points are coloured by species-specific minor clade. An ENC of 61 denotes no tendency to prefer one synonymous codon over another, whereas an ENC of 20 denotes strict use of one codon per amino acid. CFB = Chinese ferret badger.

Codon usage

The effective number of codons (ENC) is a measure of bias in codon usage, where a value of 61 denotes that each of the 61 codons are represented equally within their corresponding amino acid, and a value of 20 means only one codon is used to encode each of the 20 amino acids.

The mean ENC used by the rabies virus N gene was calculated to be 54.0, suggesting little bias in codon usage, but this varied between host-specific minor clades (Figure 1 inset). The dog-associated Cosmopolitan AF1b clade was the most biased in its codon usage (mean ENC of 51.8), with the big brown bat and arctic fox-associated clades (Bat EF-E2 and Arctic A, respectively) also being more biased than other clades. The Chinese ferret badger-associated clade Asian SEA2b was the least biased (mean ENC of 57.6).

Certain amino acids appear to have stronger codon biases than others (Figure 2; Supplementary Table S2); the arginine (R) codon AGA has an RSCU value above 2 in every host-species-specific clade investigated (RSCU values above 1.6 suggest significant overrepresentation), and was the only codon that was consistently preferred in all of the host-species-specific minor clades investigated. For every other amino acid, the most commonly preferred codon varied between clades. Only five amino acids (phenylalanine, isoleucine, asparagine, aspartic acid and glutamic acid) had no codons that were substantially over- or underrepresented (RSCU above 1.6 or below 0.6) in any clade. While preference for codons containing a UpA dinucleotide varied between codons and clades, codons containing a CpG dinucleotide were almost universally underrepresented.

PC1 values are strongly correlated with the purine content at the third codon position (adjusted R^2 of 0.729 (95%CI 0.685 - 0.772); Figure 3B). PC2 is negatively correlated with UpA content (0.500 (0.426 - 0.571); Figure 3C), and also has a weak negative correlation with CpG content (R^2 of 0.353 (0.289 - 0.418)). Multiple regression of CpG and UpA content fails to achieve a higher adjusted R^2 than UpA alone (0.500 (0.428 - 0.571)). The loadings of each codon for PC1 and PC2 are given in Supplementary Table S3.

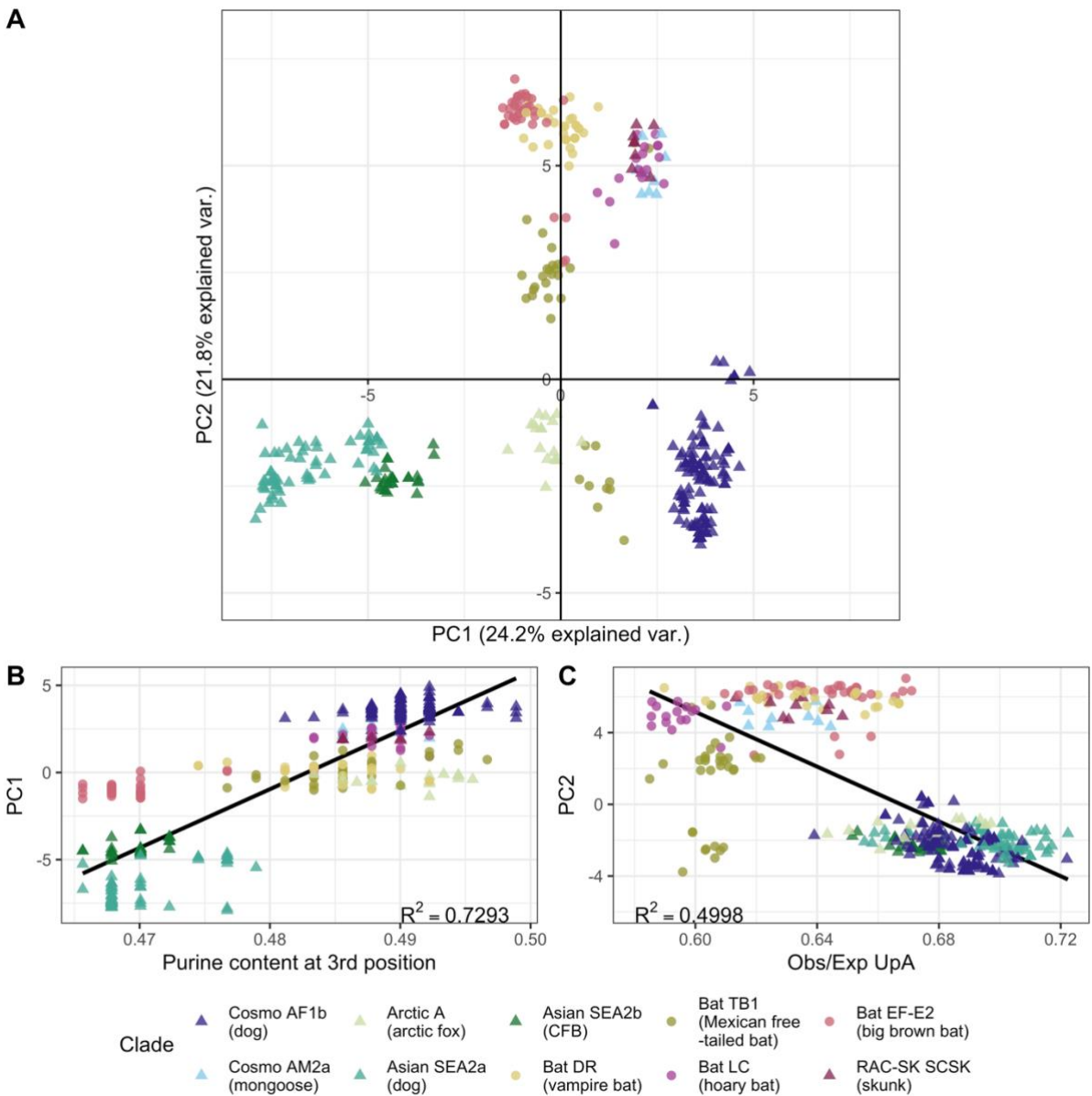


Figure 3: Codon usage between species-specific clades is different, and shaped by purine and UpA content. (A) Principal component analysis of raw codon usage values. (B) Purine

content at the third position is correlated with PC1. (C) Obs/exp UpA content is negatively correlated with PC2. The point colour represents the RABV clade. Point shape represents whether the host species is a bat (round) or a carnivore (triangle). CFB = Chinese ferret badger.

To investigate whether RABV codon usage patterns more closely resembled those of the clades' host species than its non-host species, we calculated the normalised codon adaptation index for each clade with *Canis familiaris* (domestic dog), *Vulpes lagopus* (Arctic fox), *Desmodus rotundus* (vampire bat) and *Eptesicus fuscus* (big brown bat) host codon usage values as references (Figure 4A). We did not find evidence of substantial differences across hosts in any of the RABV clades. The bat-associated RABV clades, along with the skunk-associated clade RAC-SK SCSK, had higher normalised CAI values than the carnivore-associated clades across all four host species; only the Bat EF-E2 clade, however, had a mean nCAI above 1 for all host species. nCAI was strongly correlated with CpG content (adjusted $R^2 = 0.662$ (95%CI 0.640 - 0.685); Figure 4B), implying that the higher nCAI values observed across hosts in the bat-associated clades likely due to their low CpG content, as mammalian genes are also CpG deficient (Sved & Bird 1990). UpA content was also negatively correlated with nCAI, but to a lesser extent (adjusted $R^2 = 0.610$ (0.587 - 0.631)).

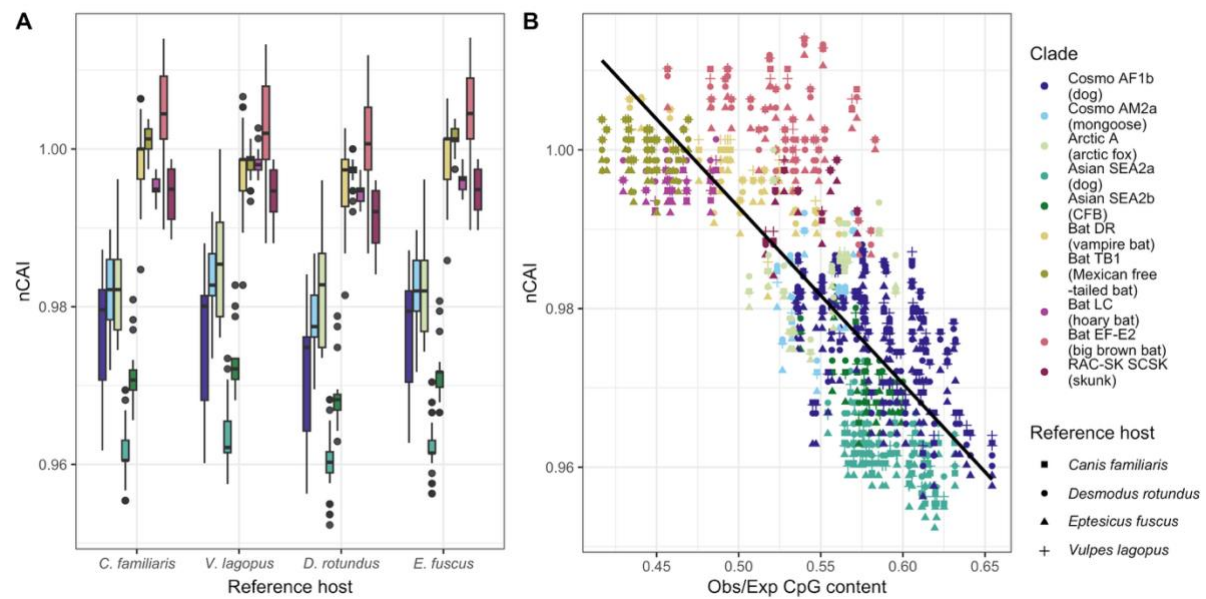


Figure 4: Normalised codon adaptation index (nCAI) values are higher in bat-associated RABV clades than in carnivore-associated clades; this is likely driven by CpG content.

(A) Codon adaptation index (CAI) of each host-specific RABV clade. *Canis familiaris* (domestic dog), *Vulpes lagopus* (arctic fox), *Desmodus rotundus* (vampire bat) and *Eptesicus fuscus* (big brown bat) codon usage values are used as reference sets; bar colour represents RABV minor clade. **(B)** nCAI is negatively correlated with CpG content. Point colour represents the host-species-specific clade, point shape represents the reference host dataset. CFB = Chinese ferret badger.

Dinucleotide content

As CpG and UpA were shown to have a large influence over RABV codon usage patterns and CAI, and are known to be selected against in RNA viruses due to pressure from immune mechanisms such as ZAP (Meagher *et al.* 2019; Odon *et al.* 2022), we further investigated the differences in these dinucleotides between the RABV clades. CpG and UpA underrepresentation (i.e., a ratio of observed-to-expected CpG or UpA content less than 1) was seen across all host-species-specific minor clades (Figure 5). Bat-associated clades had a lower

290 CpG and UpA content than carnivore-associated clades (mean observed-to-expected CpG
 291 content of 0.584 in carnivore-associated clades *versus* 0.489 in bat-associated clades; mean
 292 UpA of 0.684 in carnivore-associated clades *versus* 0.627 in bat associated clades). The Bat
 293 TB1 and Bat LC clades in particular displayed high levels of CpG suppression, whereas Bat
 294 DR and Bat EF-E2 have intermediate CpG and UpA contents, similar to that of the skunk clade
 295 RAC-SK SCSK. On the other hand, the two dog-associated clades (Cosmopolitan AF1b and
 296 Asian SEA2a), as well as the CFB-associated clade Asian SEA2b, showed the least evidence
 297 of CpG suppression. A similar pattern emerges when comparing UpA content across the clades,
 298 but with less of a clear distinction between the bat- and carnivore-associated clades than as
 299 seen for CpG.

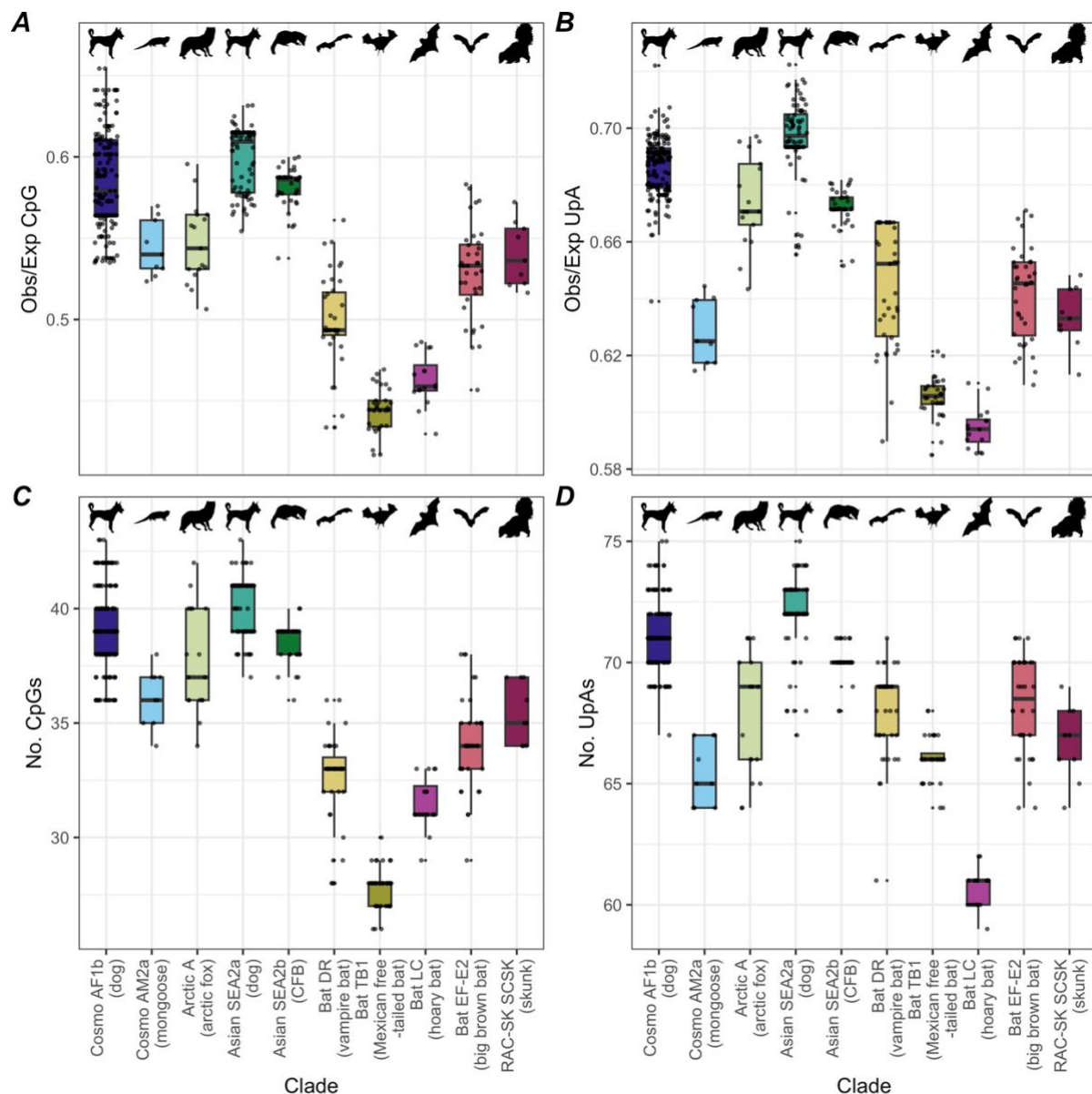


Figure 5: Ratio of observed-to-expected CpG content and GC content across host-species-specific minor clades. (A) Ratio of observed-to-expected CpG content, where a value of 1 indicates no bias, >1 indicates overrepresentation of CpG, and <1 indicates underrepresentation. **(B)** Ratio of observed-to-expected UpA content. **(C)** Number of CpG dinucleotides by clade. **(D)** Number of UpA dinucleotides by clade.

To investigate the evolutionary history of CpG and UpA content in the rabies virus, we used ancestral sequence reconstruction and calculated the CpG and UpA content of the resulting

ancestral RABV sequences. This analysis infers that the common ancestor of all the clades (pre-bat-to-carnivore shift) has a low observed-to-expected CpG content ratio (0.428), implying that low CpG is the ancestral state (Figure 6). As the common ancestor of the carnivore-associated clades has an estimated CpG content comparable to the modern carnivore-associated sequences (0.583), this suggests that higher CpG content emerged over the main bat-to-carnivore host shift. Similar patterns are observed for ancestral UpA content, with the common ancestor of all the clades having an observed-to-expected UpA ratio of 0.617, the common ancestor of the carnivore clades having a ratio of 0.704, and the common ancestor of the bat clades having a ratio of 0.638.

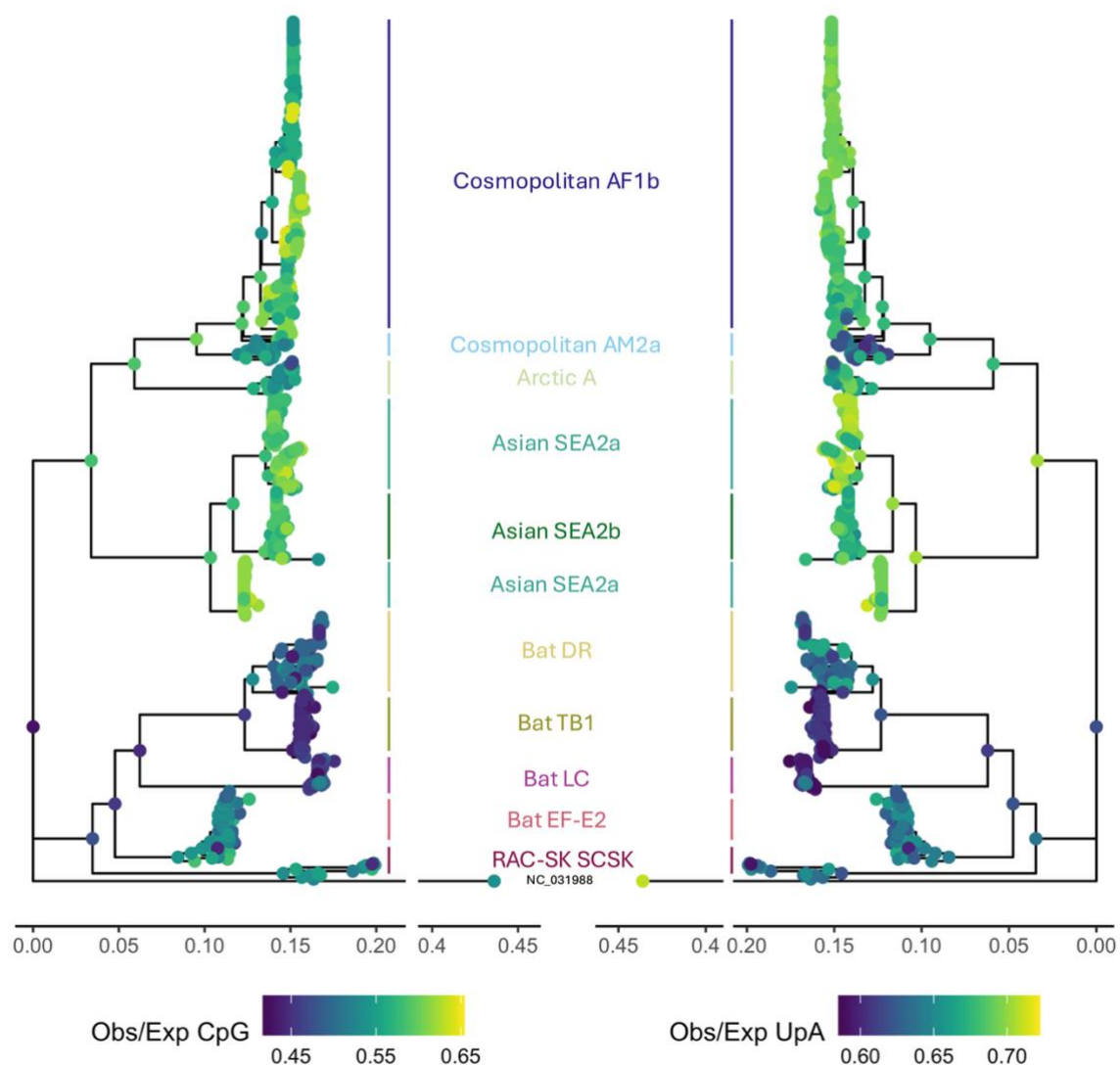


Figure 6: Low CpG and UpA content appears to be the ancestral state of RABV.

Phylogenetic tree of all the N gene sequences with ancestral node sequences reconstructed and CpG (left) and UpA (right) content calculated. Point colour represents the observed dinucleotide content as a proportion of the expected dinucleotide content given the sequence's nucleotide content. Note the break in the x-axis between 0.2 and 0.4 substitutions per site to truncate the outgroup branch.

When comparing the frequencies of mouse-ZAP's optimal binding motifs in the N gene sequences to similar but suboptimal motifs, we found that four clades (Asian SEA2b, Bat TB1, Bat DR and Bat EF-E2) had very few optimal motifs compared to their number of suboptimal motifs, suggesting effective selection against the optimal motifs by ZAP (Figure 7). Bat LC had very few binding motifs overall, with none present in the second half of the gene, but more optimal binding motifs were present than suboptimal. Cosmopolitan AF1b and Arctic A had equal or higher counts of the optimal motifs than the suboptimal motifs in every sequence, suggesting either a selection pressure against the mouse-ZAP suboptimal motifs, or towards the optimal motifs. As these optimal and suboptimal motifs occur at different loci, it is unlikely that suboptimal binding motifs were directly replaced with optimal binding motifs over the bat-to-carnivore host shift.

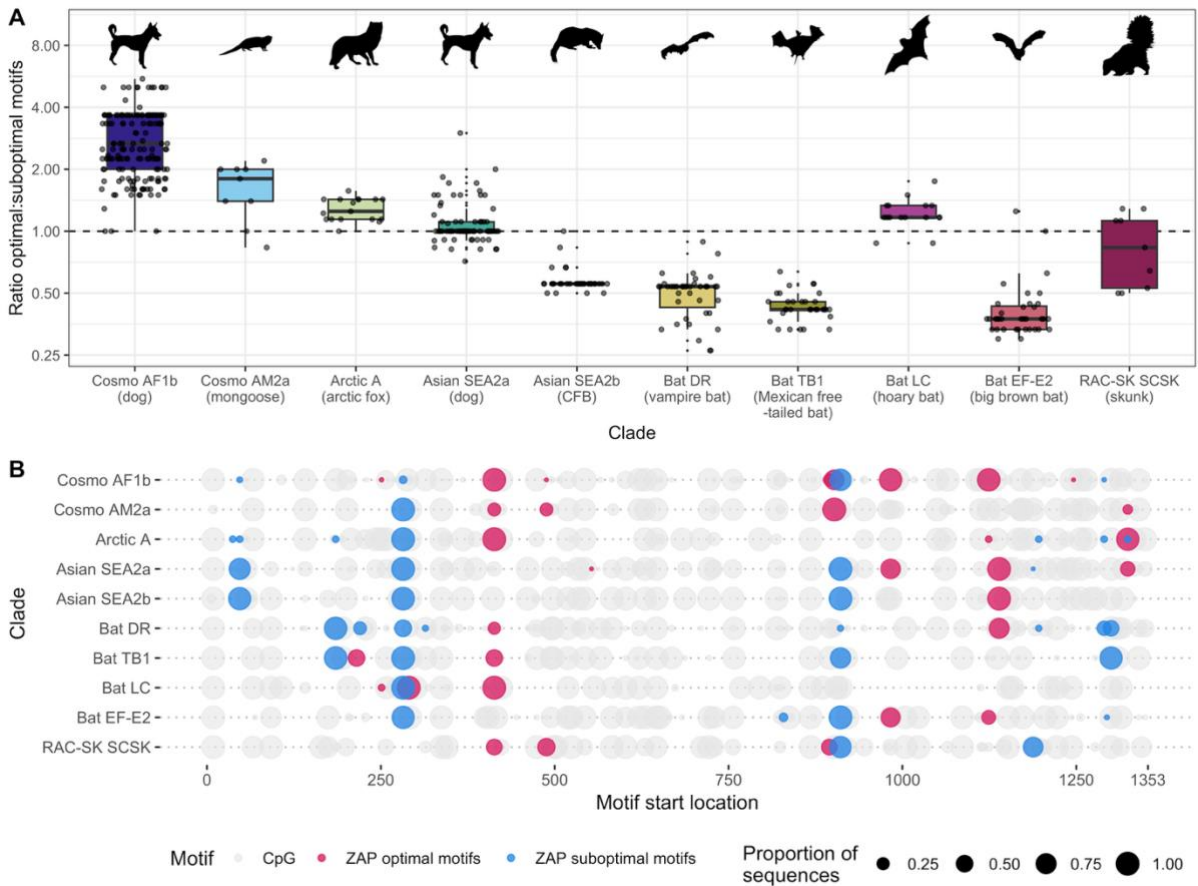


Figure 7: N gene sequences from all bat-associated clades except Bat LC on average contain more ZAP suboptimal binding motifs than optimal motifs, whereas some

carnivore clades contain more optimal binding motifs. (A) Ratio between the number of optimal and suboptimal ZAP binding motifs by species-specific RABV clade. Note that the y-axis is log scaled. (B) Locations of CpG dinucleotides (grey), ZAP optimal motifs (pink) and ZAP suboptimal motifs (blue) on the N gene show that the acquired CpG dinucleotides in carnivore-associated clades occur both within and outside of ZAP-optimal motifs. Point size represents the proportion of sequences within each clade where each motif is present at each locus.

Discussion

We found that much of the variation in codon usage between host-species-specific rabies virus clades can be explained by purine content at the third codon position and UpA content. We also found that CpG content is significantly lower in bat-associated RABV clades than in carnivore-associated clades. Together with the over-representation of ZAP optimal binding motifs in some carnivore-associated clades, this may indicate differences in the mechanism of ZAP between the host species, or in the exposure to ZAP between the clades.

The results of our principal component analysis indicated that most host-species-specific rabies virus clades had distinct patterns of codon usage, whereas the RAC-SK south-central skunk, Cosmopolitan AM2a and Bat LC clades clustered closely together, despite belonging to different major clades and infecting hosts belonging to different taxonomic families. While the bat- and carnivore-associated clades appear to broadly split across PC2, the two dog-associated clades are the most distant clades across PC1. Further analysis revealed that PC1 is significantly correlated with purine content at the third codon position, which explains 73% of the variation along PC1. Higher purine content in the virus' genomic RNA, which would correspond to a

lower purine (or higher pyrimidine) content in the coding RNA sequences used in our analysis, is associated with restricted access of polymerase to the genomic RNA through the nucleocapsid, and thus reduced transcription (Gumpper *et al.* 2019). It is currently unclear whether the modulation of transcription or another factor drives the differences in purine content at the third codon position between RABV clades; as the two dog-associated clades have very different purine contents, it seems unlikely that these differences are purely due to host adaptation. As little is currently known about the role of purine and pyrimidine content in negative-strand RNA viruses, further work on the topic could reveal the evolutionary forces driving the differences between these clades.

PC2, on the other hand, is correlated with UpA content ($R^2 = 0.50$), and to a lesser extent CpG content ($R^2 = 0.35$). CpG content in RNA viruses is almost universally lower than would be expected based on GC content due to selective pressure from the host's zinc-finger antiviral protein (ZAP) which specifically targets motifs containing CpG dinucleotides (Takata *et al.* 2017). UpA, another target of ZAP, and is also depleted in RNA virus genomes, but to a lesser extent than CpG (Odon *et al.* 2022). Bat-associated RABV clades have a CpG content that would be considered typical for negative-strand RNA viruses, while the carnivore-associated clades have noticeably higher CpG content, though still within the known range for mammalian negative-strand RNA viruses (Simmonds *et al.* 2013). A similar, but less clear, pattern is observed for UpA; this is somewhat surprising in light of our finding that UpA content has more influence over PC1 than CpG content. Carnivore host genes generally do not appear to have a higher CpG content than bat genes (Shaw *et al.* 2021), so the significant difference in CpG content between bat- and carnivore-associated clades is not likely to be due to host mimicry. This finding is also opposite to a similar analysis conducted in coronaviruses, where coronaviruses with bat hosts had a higher CpG content than coronaviruses found in other

mammals (MacLean *et al.* 2021; Nchioua *et al.* 2020). This may point to a RABV-specific reduction in exposure to ZAP in clades infecting carnivores as opposed to bats. We also found that some carnivore-associated RABV clades had more CpG-based ZAP optimal binding motifs (in the format C(n_m)G(n)CG, as identified in mice) than the negative control suboptimal motifs, whereas most of the bat-associated clades had more suboptimal motifs than optimal, as would be expected under the hypothesis that RABV has evolved to avoid being targeted by ZAP. This may further support the possibility of RABV being less exposed to ZAP in carnivore hosts than in bat hosts, therefore easing the selection pressure against CpG and the optimal binding motif. If this is true, we would expect the skunk-associated clade's CpG content to continue to increase; while it has higher mean CpG content than the bat-associated clades, it is not yet as high as the other carnivore-associated clades.

Another explanation for the variation in the number of ZAP optimal and suboptimal motifs between clades is that the optimal binding motif of ZAP may differ across RABV's broad range of host species; this could explain differences in binding-motif ratios without a difference in selection pressure from ZAP. This could also affect CpG content if the hosts' ZAP optimal motifs differ on expected CpG content. It is also possible that other immune mechanisms or selective pressures may be promoting a high-CpG state in carnivore-associated rabies clades that outweigh the deleterious effects of increased ZAP targeting. When considering the bat-associated clades only, CpG content does not appear to correlate with host phylogeny, implying that non-genetic host factors (such as migration habits, which have been proposed to alter bats' immune responses (Becker *et al.* 2025; Rogers *et al.* 2022; Voigt *et al.* 2020)) may affect the virus' exposure to ZAP, and thus selection against CpG content. Further study is required to elucidate what mechanisms may have caused the easing of selection pressure against CpG and UpA content during the bat-to-carnivore host shift, such as potential mechanistic differences

(e.g., optimal binding motifs) between bat- and carnivore- derived ZAP and its associated proteins, or differences in the probability of the exposure of the virus to ZAP, and why this appears to be stronger in some bat-associated clades than others. More information about species-specific CpG- and UpA-based ZAP binding motifs may shed more light on this.

The CAI analysis suggests that RABV codon usage is not evolving to match the codon usage of specific hosts. Our findings instead suggest that bat-associated RABV codon usage more closely resembles both carnivore- and bat-hosts' codon usage than carnivore-associated RABV clades do, but that RABV codon usage overall is not more similar to either bat or carnivore codon usage. It seems likely that the higher CAI values observed in the bat-associated RABV clades are simply reflective of their lower CpG content, as vertebrate genes also have low CpG content due to the deamination of methylated CpG sites (Sved & Bird 1990). As CoCoPUTs' codon usage data for non-human animals cannot be filtered for genes associated with certain tissue types, it is possible that the CAI analysis could be affected by the high tissue-specificity of codon usage in multicellular organisms (Payne & Alvarez-Ponce 2019; Plotkin *et al.* 2004), especially if some host species' codon usage data are biased towards certain tissue types.

One limitation of our investigation is that we focused only on ten minor RABV clades with clear species associations, leaving some major clades unrepresented. Another is that we analyzed only the N gene. This was chosen as it was the most commonly sequenced gene, providing the largest number of sequences across the clades, but other genes may be subject to different selective pressures. Further investigation is needed to confirm whether the patterns observed in our study are consistent across the genome, and in other lyssaviruses.

We found differences in the codon usage between host-species-specific RABV clades, a large proportion of which were driven by UpA content and purine content at the third codon position. CpG content was also significantly higher in carnivore-associated clades than in bat-associated clades, which may suggest underlying differences in immune mechanisms between the host groups, and should be explored further. These results may also have implications for the efficacy of codon-deoptimised or CpG-enriched live attenuated vaccines across different host-species-specific clades should they be developed for the rabies virus, as they have been for Influenza A (Sharp *et al.* 2023).

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Author contributions

RD: Conceptualisation, data curation, formal analysis, investigation, methodology, visualisation, writing - original draft, writing - review and editing

JD: Supervision, methodology, writing - review and editing

MA: Writing - review and editing

CC: Supervision, writing - review and editing

KH: Supervision, funding acquisition, writing - review and editing

Data availability

All of the code and data used in this study are available in the GitHub repository <https://github.com/RowanDurrant/rabies-codon-bias>. The accession numbers of all sequences used are available in Supplementary Table S1.

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Supplementary material

Supplementary Table S1: Accession numbers for the sequences used in our analysis by host species.

Clade (Host)	Accession numbers
Asian SEA2b (Chinese ferret badger)	FJ598135, FJ712195, FJ712196, GU647092, KF726852, KP319184, KP319185, KP319186, KP319187, KP319188, KP319189, KP319190, KP319191, KP319192, KP319193, KP319194, KP319195, KP319196, KP319197, KP319198, KP319199, KP319200, KP319201, KP319202, KP319203, KP319204, KP319205, KP319206, KP319207, KP319208, KP319209, KP319210, KP319211, KP319212, KP319213
Asian SEA2a (dog)	DQ666289, DQ666290, DQ866105, DQ866106, DQ866108, DQ866109, DQ866110, DQ866112, DQ866116, DQ866119, EU086182, EU086183, EU159386, EU828653, FJ561726, FJ561727, FJ561728, FJ866829, FJ866830, FJ866831, GQ472468, GQ472470, GQ472473, GQ472477, GU358653, HM486355, HM486356, HM486357, HM486358, HM486359, HM486361, HM486362, HM486364, HM486365, HM486366, HM486367, HM486368, HM486369, HM486370, HM486371, HM486372, HM486373, HM486374, HM486375, HM486379, HM486380, HM486381, JF819605, JF819615, JF819622, JN974823, JN974824, JN974828, JN974829, JN974838, JN974842, JN974845, JN974854, JN974870, JN974871, JN974872, JN974873, JQ970480, KT221107, KT894558, KT894572, KT894573, KT894577, KT894578, KX148264, KY451767, MG201879, MG201880, MG201884, MG201919, MG201921
Arctic A (Arctic fox)	AY352488, AY352501, KX954123, LT598537, LT598538, LT598539, LT598540, LT598541, LT598542, LT598543, LT909550, MN384717, AY352487, EF611828, EF611830, EF611851, L20675
Cosmopolitan AF1b (dog)	AB284509, AB284510, AB284511, AB284512, AB284513, AB284514, AB285215, EU853582, EU853584, EU853585, EU853586, EU853587, EU853588, EU853589, EU853590, HM179504, HM179505, HM179506, KF155002, KR534217, KR534218, KR534219, KR534220, KR534228, KR534229, KR534230, KR534231, KR534233, KR534238, KR534244, KR534247, KR534249, KR534250, KR534251, KR534252, KR534253, KR906734, KR906735, KR906736, KR906739, KR906741, KR906742, KR906743, KR906744, KR906745, KR906746, KR906747, KR906748, KR906749, KR906750, KR906751, KR906752, KR906754, KR906755, KR906756, KR906757, KR906759, KR906760, KR906762, KR906763, KR906765, KR906766, KR906767, KR906768, KR906770, KR906771, KR906772, KR906773, KR906774, KR906776, KR906780, KR906782, KR906784, KR906789, KR906790, KR906792, KT336432, KT336433, KT336434, KT336435, KT336436, KT336437, KX148203, KX148204, KX148205, KX148206, KX148208, KY210222, KY210223, KY210224, KY210226, KY210227, KY210229, KY210231, KY210232, KY210234, KY210235, KY210238, KY210239, KY210243,

	KY210244, KY210245, KY210247, KY210248, KY210249, KY210250, KY210251, KY210252, KY210253, KY210254, KY210255, KY210256, KY210257, KY210258, KY210260, KY210263, KY210264, KY210265, KY210266, KY210267, KY210269, KY210270, KY210271, KY210272, KY210273, KY210278, KY210279, KY210280, KY210281, KY210282, KY210286, KY210287, KY210288, KY210289, KY210290, KY210292, KY210293, KY210294, KY210295, KY210296, KY210299, KY210300, KY210303, KY210304, KY210307, KY210311, KY563717, LC029890, MN726804, MN726826, MN726827, MT454631, MT454634, MT454635, MT454636, MT454637, MT454639, MT454641, MT454643, MT454644, MT454645
Bat LC (hoary bat)	AF351845, AF351846, AF351858, AF394883, AF394884, GU644712, GU644713, GU644714, GU644715, GU644716, GU644717, GU644718, GU644719, GU644720, GU644721, JQ685947
Cosmopolitan AM2a (mongoose)	AY854505, AY854525, AY854531, AY854544, AY854546, AY854558, AY854567, AY854573, AY854576
RAC-SK SCSK (skunk)	EU345002, EU345003, EU345004, JQ685938, JQ685968, MW055084, MW055085, MW055086, MW055087
Bat DR (vampire bat)	AB519642, AF070449, AF351847, AF351852, AY854587, AY854592, AY854594, AY877433, AY877434, AY877435, GU592648, KF656696, KF656697, KF864234, KF864322, KF864397, KM594040, KM594041, KM594042, KP202393, KT023101, KU523255, MN968374, MN968375, MN968376, MN968377, MN968378, MN968379, MN968380, MN968381, MN968382, MN968383, MN968384, MN968385, MN968386, MN968387, MT891038, MW249020, MW249021
Bat EF-E2 (big brown bat)	AF351828, AF351831, AF351832, AF351833, AF351855, AF351861, AF351862, AY039227, AY039228, AY039229, GU644652, GU644654, GU644655, GU644656, GU644657, GU644658, GU644659, GU644660, GU644661, GU644662, GU644663, GU644664, GU644665, GU644666, GU644667, GU644668, GU644669, GU644670, GU644671, GU644676, GU644677, GU644684, GU644689, GU644690, GU644695, JQ685925
Bat TB1 (Mexican free-tailed bat)	AF351849, AF394876, GU644760, GU644761, GU644762, GU644763, GU644764, GU644765, GU644766, GU644767, GU644768, GU644769, GU644770, GU644771, GU644772, GU644773, GU644774, GU644775, GU644776, GU644777, GU644778, GU644779, GU644780, GU644781, GU644782, GU644783, GU644784, GU644785, GU644786, GU644787, GU644788, JQ685905

Supplementary Table S2: Mean RSCU values across different clades. Green highlight indicates that the codon is significantly underrepresented (RSCU value < 0.6), orange highlight indicates that the codon is significantly overrepresented (RSCU value > 1.6). Bold indicates the most preferred codon for each amino acid in each clade.

Amino acid	Codon	Asian SEA2b (CFB) (n = 35)	Cosmo AF1b (dog) (n = 161)	Asian SEA2a (dog) (n = 76)	Arctic A (Arctic fox) (n = 12)	Cosmo AM2a (mongoose) (n = 9)	RAC-SK SCSK (skunk) (n = 9)	Bat LC (hoary bat) (n = 16)	Bat EF-E2 (big brown bat) (n = 36)	Bat TB1 (Mexican free tailed bat) (n = 32)	Bat DR (vampire bat) (n = 39)
Phe	UUU	0.885	0.636	0.819	0.772	0.642	1.01	0.972	1.07	1.12	0.867
	UUC	1.12	1.36	1.18	1.23	1.36	0.990	1.03	0.932	0.877	1.13
Leu	UUA	1.02	0.525	1.12	0.486	0.723	0.615	0.903	1.04	0.965	1.08
	UUG	1.24	1.26	1.07	1.66	1.50	1.77	2.11	1.80	1.63	1.54
	CUU	0.886	0.352	0.968	0.571	0.332	0.938	0.859	0.607	0.868	0.682
	CUC	0.719	0.540	0.513	0.528	0.704	0.693	0.528	0.709	0.519	0.678
	CUA	1.06	1.72	1.00	0.929	1.29	0.724	0.704	0.522	1.43	0.594
	CUG	1.07	1.60	1.33	1.83	1.45	1.26	0.891	1.40	1.43	1.42
Ile	AUU	1.15	1.01	1.12	1.06	1.07	0.948	1.03	0.853	1.04	1.09

	AUC	0.740	0.889	0.837	0.935	0.915	0.842	0.858	1.03	1.04	0.871
	AUA	1.11	1.10	1.04	1.00	1.02	1.21	1.11	1.11	0.921	1.04
Val	GUU	1.17	1.01	0.983	1.07	0.989	0.983	1.28	1.07	1.22	1.16
	GUC	1.34	1.43	1.39	1.17	1.46	1.09	1.00	0.737	0.932	0.987
	GUA	0.669	0.574	0.958	0.862	0.548	0.637	0.711	0.752	0.634	0.732
	GUG	0.812	0.984	0.667	0.899	1.01	1.29	1.00	1.44	1.21	1.12
Ser	UCU	1.19	1.27	1.61	1.24	1.18	1.33	1.41	1.70	1.73	1.52
	UCC	1.25	0.809	0.922	0.827	0.895	1.47	0.640	0.701	0.783	0.863
	UCA	1.61	1.83	1.28	1.72	1.89	1.22	1.74	1.82	1.58	1.53
	UCG	0.487	0.763	0.599	0.656	0.667	0.646	0.305	0.388	0.320	0.418
	AGU	0.980	0.982	1.04	0.826	1.05	0.677	1.27	0.894	0.955	0.972
	AGC	0.483	0.348	0.554	0.727	0.324	0.662	0.640	0.497	0.631	0.701
Pro	CCU	1.24	1.57	1.04	1.63	1.42	1.02	1.63	1.62	1.77	1.94
	CCC	1.00	0.916	1.21	1.22	1.08	0.905	0.735	0.720	0.984	0.541

	CCA	1.02	0.909	0.901	1.16	1.47	0.971	0.941	0.537	0.492	0.521
	CCG	0.736	0.601	0.850	0	0.0278	1.11	0.691	1.13	0.758	0.995
Thr	ACU	1.37	1.15	1.49	1.49	1.35	1.33	0.978	0.936	1.42	1.31
	ACC	0.971	1.25	0.907	0.756	0.837	1.00	1.37	1.36	0.777	0.816
	ACA	1.22	1.45	0.987	0.678	1.49	1.45	1.51	1.41	1.55	1.63
	ACG	0.438	0.160	0.615	1.08	0.320	0.210	0.139	0.268	0.258	0.251
Ala	GCU	1.28	0.968	1.44	1.36	1.06	1.45	0.852	1.13	1.48	1.35
	GCC	1.08	1.09	0.833	0.851	0.990	0.730	0.982	0.986	0.883	0.846
	GCA	1.50	1.72	1.49	1.66	1.69	1.74	1.80	1.84	1.15	1.42
	GCG	0.143	0.216	0.239	0.132	0.257	0.0839	0.363	0.0443	0.480	0.392
Tyr	UAU	1.07	1.25	1.16	1.48	1.28	1.25	0.851	1.41	1.33	1.20
	UAC	0.930	0.748	0.836	0.516	0.720	0.748	1.15	0.592	0.670	0.796
His	CAU	0.519	1.22	0.810	0.923	1.27	1.08	1.23	1.23	1.30	1.28
	CAC	1.48	0.780	1.19	1.08	0.735	0.918	0.769	0.771	0.700	0.722

Gln	CAA	1.19	1.04	1.53	1.20	1.00	0.728	1.01	1.06	1.12	0.810
	CAG	0.806	0.963	0.469	0.800	1.00	1.27	0.988	0.941	0.879	1.19
Asn	AAU	0.955	1.08	0.953	1.14	0.883	1.09	0.918	1.33	1.24	1.27
	AAC	1.05	0.924	1.05	0.862	1.12	0.914	1.08	0.671	0.764	0.726
Lys	AAA	1.06	0.914	0.793	0.479	0.885	0.703	0.589	0.678	0.960	0.955
	AAG	0.943	1.09	1.21	1.52	1.11	1.30	1.41	1.32	1.04	1.05
Asp	GAU	1.02	0.997	0.772	1.04	0.992	0.982	0.867	0.900	0.956	0.956
	GAC	0.977	1.00	1.23	0.964	1.01	1.02	1.13	1.10	1.04	1.04
Glu	GAA	0.932	0.880	0.842	0.683	0.805	1.20	0.833	0.768	0.920	0.841
	GAG	1.07	1.12	1.16	1.32	1.20	0.805	1.17	1.23	1.08	1.16
Cys	UGU	1.02	1.97	0.610	1.36	1.43	1.13	1.06	1.01	1.33	1.02
	UGC	0.981	0.0284	1.39	0.639	0.571	0.866	0.938	0.985	0.667	0.978
Arg	CGU	0.781	0.804	0.781	0.786	0.746	0.495	0.480	0.531	0.504	0.360
	CGC	0	0.0153	0	0	0.0303	0.522	0.240	0.225	0.252	0.403

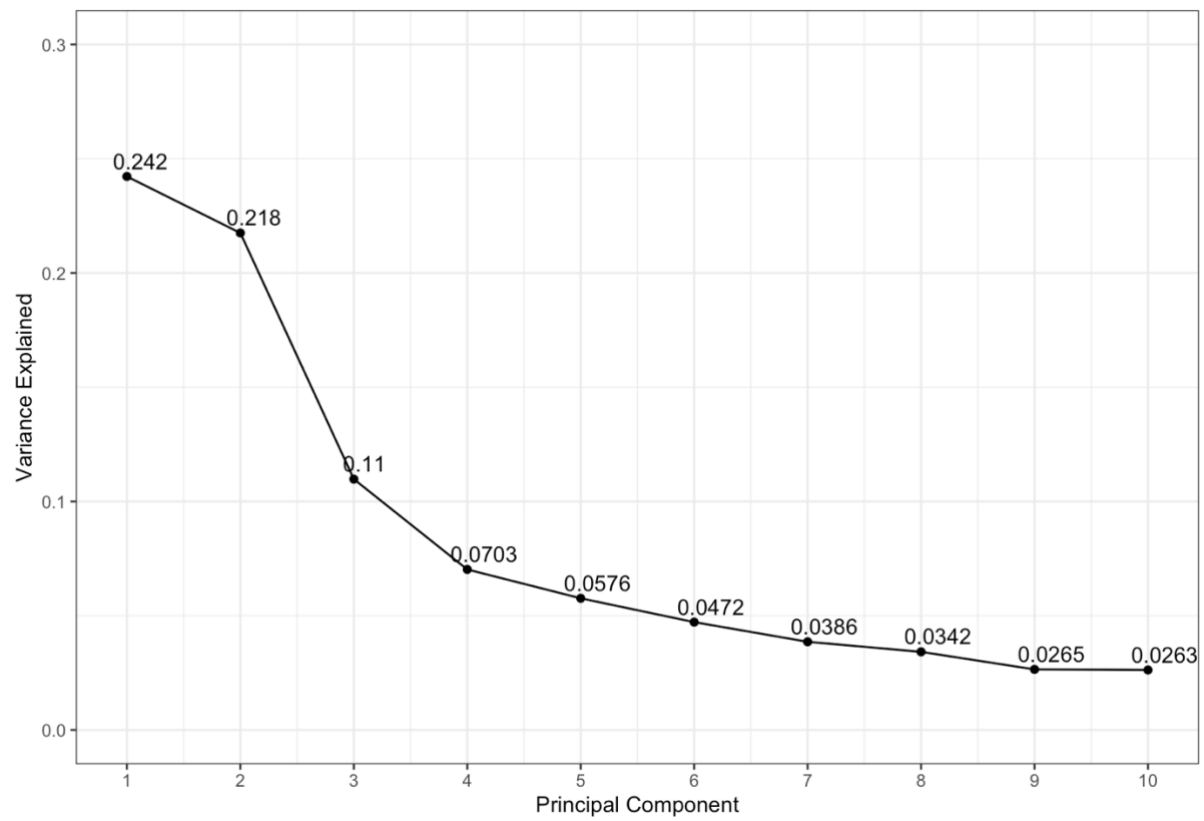
	CGA	0.781	0.368	0.777	0.262	1.06	0.378	0.480	0.0290	0	0.276
	CGG	0.260	0.443	0.265	0.786	0.317	0.520	0.495	0.792	0.512	0.399
	AGA	2.32	3.11	2.76	3.27	2.84	2.48	2.42	3.62	2.76	2.79
	AGG	1.86	1.26	1.42	0.893	1.00	1.61	1.89	0.807	1.98	1.77
Gly	GGU	1.07	0.756	0.886	0.763	0.838	0.894	0.699	0.940	0.556	0.432
	GGC	0.401	0.716	0.548	0.452	0.603	0.773	0.949	0.529	0.556	0.934
	GGA	1.15	1.04	1.27	1.02	1.38	1.20	0.975	1.20	1.66	1.62
	GGG	1.38	1.49	1.30	1.76	1.18	1.13	1.38	1.33	0.966	1.01

Supplementary Table S3: PCA loadings for each codon on PC1 and PC2.

Amino acid	Codon	PC1	PC2
F	TTT	-0.103	0.203
	TTC	0.106	-2.03
L	TTA	-0.196	0.123
	TTG	0.0554	0.188
	CTT	-0.240	0.0814
	CTC	-0.0153	0.114
	CTA	0.123	-0.226
	CTG	0.122	-0.0441
I	ATT	-0.0568	-0.0821
	ATC	0.121	0.0894
	ATA	0.133	-0.0953
V	GTT	-0.0394	0.0690
	GTC	-0.00970	-0.247
	GTA	-0.198	-0.00404
	GTG	0.151	0.166
S	TCT	-0.109	0.176
	TCC	-0.124	-0.0386
	TCA	0.193	0.0143
	TCG	0.0713	-0.192
	AGT	-0.0850	-0.00106

	AGC	-0.124	0.162
P	CCT	0.160	0.152
	CCC	-0.121	-0.142
	CCA	-0.0174	-0.174
	CCG	-0.0441	0.146
T	ACT	-0.173	0.0352
	ACC	0.163	-0.0262
	ACA	0.139	0.157
	ACG	-0.170	-0.0290
A	GCT	-0.176	0.0874
	GCC	0.148	-0.116
	GCA	0.159	-0.125
	GCG	-0.00671	0.0895
Y	TAT	0.105	0.0607
	TAC	-0.104	-0.0607
H	CAT	0.194	0.122
	CAC	-0.206	-0.106
Q	CAA	-0.186	-0.128
	CAG	0.187	0.0763
N	AAT	0.0680	0.188
	AAC	-0.121	-0.189

K	AAA	0.0419	-0.0592
	AAG	0.0201	0.00920
D	GAT	0.123	-0.00413
	GAC	-0.175	0.00257
E	GAA	0.0360	-0.0309
	GAG	0.0209	-0.0382
C	TGT	0.228	-0.106
	TGC	-0.229	0.0898
R	CGT	-0.0177	-0.246
	CGC	0.0184	0.236
	CGA	-0.157	-0.168
	CGG	0.102	0.133
	AGA	0.113	0.0752
	AGG	-0.0998	0.0730
G	GGT	-0.0710	-0.122
	GGC	0.116	0.0962
	GGA	-0.0756	0.174
	GGG	0.0673	-0.185



Supplementary Figure S1: Scree plot for the PCA of raw codon usage values.